



**FIRST SEMESTER, 2026–2027**  
**Course Handout Part II**

Date: 30th July 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

**Course No** : CS G526  
**Course Title** : Advanced Algorithms & Complexity  
**Instructor-in-Charge** : Tulasimohan Molli (TA: Hans Krupakar)  
**Instructors** : Tulasimohan Molli

## 1. Course Description:

This course explores advanced paradigms in algorithm design and analysis beyond the core curriculum: randomized algorithms, approximation algorithms, game-theoretic techniques, and parallel & distributed algorithms, with applications to modern domains such as the Internet/Web and computational biology.

## 2. Scope & Objective of the Course:

The course equips students with the mathematical tools and algorithmic techniques needed to reason about computationally hard problems and to design efficient algorithms under uncertainty. Emphasis is placed on both rigorous theoretical analysis and practical algorithm engineering.

## 3. Textbooks:

- T1.** *Randomized Algorithms*, R. Motwani & P. Raghavan, CUP, 1995.
- T2.** *Combinatorial Optimization: Algorithms and Complexity*, C. H. Papadimitriou & K. Steiglitz, PHI, 1982.

## References:

- R1.** *The Design of Approximation Algorithms*, D. Williamson & D. Shmoys, Cambridge University Press.
- R2.** *Design and Analysis of Randomized Algorithms*, J. Hromkovic, Springer.
- R3.** *Approximation Algorithms*, V. Vazirani, Springer.
- R4.** *Complexity and Approximation*, G. Ausiello, P. Crescenzi, G. Gambosi, V. Kann, A. Marchetti-Spaccamela, M. Protasi, Springer.
- R5.** *Algorithm Design*, J. Kleinberg & E. Tardos, Pearson Education.
- R6.** Additional reading assigned by the Instructor.

## 4. Course Plan:

Weekly plan for 14 weeks with 3 lectures per week

| Week no. | Module No. | Description                                                                                 | Reference |
|----------|------------|---------------------------------------------------------------------------------------------|-----------|
| 1.       | M1         | Introduction — Advanced Algorithms & Complexity; Review of Probability & Complexity Classes | T1 C1     |
| 2.       | M1         | Las Vegas & Monte Carlo, Freivalds, Min-Cut; PIT, Schwartz–Zippel                           | T1 C1     |
| 3.       | M2         | Tail Inequalities, Chebyshev, Chernoff Bounds and Applications                              | T1 C3–4   |
| 4.       | M3         | Skip Lists, Hash Tables, Cuckoo Hashing                                                     | T1 C8     |
| 5.       | M3         | Randomized Graph Algorithms: MST, Shortest Paths, Matching                                  | T1 C10    |
| 6.       | M4         | Game Theory: Minimax, Zero-Sum, Equilibrium, Multiplicative Weights                         | T1 C4     |
| 7.       | M5         | Competitive Ratio, Ski Rental, Paging, Secretary Problem                                    | T1 C13    |
| 8.       | M6         | Euclid, Euler's phi, Primality Testing                                                      | T1 C14    |
| 9.       | M7         | PRAM, Prefix, MIS; Byzantine Agreement, Consensus                                           | T1 C12    |
| 10.      | M8         | Tutte Matrix, Perfect Matchings, Isolation Lemma, #P & Counting                             | T1 C1, AR |
| 11.      | M9         | Greedy, LP Rounding, Primal-Dual, Local Search; Set Cover, TSP                              | R1, R3    |
| 12.      | M9         | Approximation Schemes: PTAS, APX; Inapproximability, Gap Technique                          | R4        |
| 13.      | M10        | PageRank, Web Graph, Recommendation Systems; Sequence Alignment                             | AR        |
| 14.      | M11        | Student Project Proposal Presentations                                                      | —         |

## 5. Evaluation Scheme:

| Component                 | Wt (%) | Duration   | Date & Time | Comments                                          |
|---------------------------|--------|------------|-------------|---------------------------------------------------|
| Class participation       | 10     | in class   | throughout  | Open Book                                         |
| Term Project              | 20     | during sem | throughout  | proposal + final presentation + report, Open Book |
| Lab participation         | 10     | weekly     | Wed/Fri     | Weekly problem solving exercises, Open Book       |
| Mid-Semester              | 25     | 90 mins.   | 06/10       | Closed Book                                       |
| Comprehensive Examination | 35     | 180 mins.  | 04/12 AN    | Closed Book                                       |

**NC Criterion:** 30% of the average of the top three performers or 40% of median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department.

## 6. Office Consultation Hour:

Monday 3:00–4:00 PM, H-134. Also, feel free to email me to seek a prior appointment.

## **7. Make-up Policy:**

Make-up tests will be given only to genuine cases with the prior permission of the Instructor-in-Charge, following the regulations set by the AGSRD office. No make-up will be given for the tutorial tests.

## **8. Academic Honesty and Integrity Policy:**

Academic honesty and integrity are to be maintained by all the students throughout the semester and no type of academic dishonesty is acceptable.

## **9. Disclaimer:**

Any kind of unfair means in exams and assignments will be strictly dealt.



**Instructor-in-charge  
CS G526: Advanced Algorithms & Complexity**